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One-pot multi-step cascade protocols toward β-indolyl sulfoximidoyl amides *via* intermolecular trapping of an α-indolylpalladium complex by CO†

Huahua Liu, Li Wei and Zhiyuan Chen *

Various β-indolyl sulfoximidoyl amides were efficiently prepared from *ortho*-iodoanilines, propargyl bromides, 1 atm of CO, and substituted *N*H-sulfoximines, through a palladium-catalyzed indole annulation/carbonyl insertion/C–N bond formation cascade. Mostly good to high yields of the products were obtained through this multi-step, one-pot reaction protocol under very gentle reaction conditions. The obtained β-indolyl sulfoximidoyl amides could be converted into biologically interesting sulfoximine analogues that contain a tryptamine moiety.

Introduction

Sulfoximines represent an important class of structural motifs that are found in natural products, pharmaceuticals and agrochemicals.[1] Some typical sulfoximine-containing molecules, such as Suloxifen from Gödecke GmbH,[2] pan-CDK inhibitor Roniciclib (BAY 1000394) from Bayer,[3] and AZD6738 from AstraZeneca,[4] have found important medicinal applications in practical areas (Fig. 1). In a recent study, Sirvent and Lücking from Bayer evaluated sulfoximine analogues as an isostere for amines by incorporation into marketed drugs and advanced clinical candidates with comparison of the biological properties.[5] Bolm and co-workers demonstrated the chemical degradation transformations in the laboratory leading to sulfoximine degradations.[6] Besides, sulfoximines can also act as directing groups in C–H activations,[7] or as chiral auxiliaries or ligands in transition-metal mediated asymmetric catalysis.[8] Due to these reasons, studies on the synthetic developments of sulfoximine chemistry have aroused widespread interest in the past few decades.[1] On the other hand, the tryptamine scaffold, possessing an ethanamine moiety at the C-3 position of indole,[9] was recognized as a "privileged structure" that has been found in psychoactive compounds like serotonin and amphetamine (Fig. 1).[10] Compounds bearing a 3-(2-aminoethyl)-1*H*-indole moiety usually exhibit a high agonist action at multiple receptors.[11] Therefore, attempts to construct diversely functionalized tryptamine libraries proved to be interesting in drug development or in medicinal research.[12] We envisioned that the installation of a sulfoximine moiety at the *N* terminus of the tryptamine scaffold to form β-indoly sulfoximidoyl amide **A** should be beneficial for subsequent biological assays (Fig. 1).

Synthetic approaches toward functionalized sulfoximines usually start from free *N*H-sulfoximines. To date, divergent transformations of the sulfoximine N–H bond have enabled N–C bond formation reactions *via* versatile synthetic methods, including alkylation,[13] vinylation,[14] alkynylation,[15] arylation,[16] propargylation,[17] and cyanation.[18] However, these methods are not applicable for the rapid production of acylated sulfoximines. *N*-Acylation of sulfoximines is generally performed by

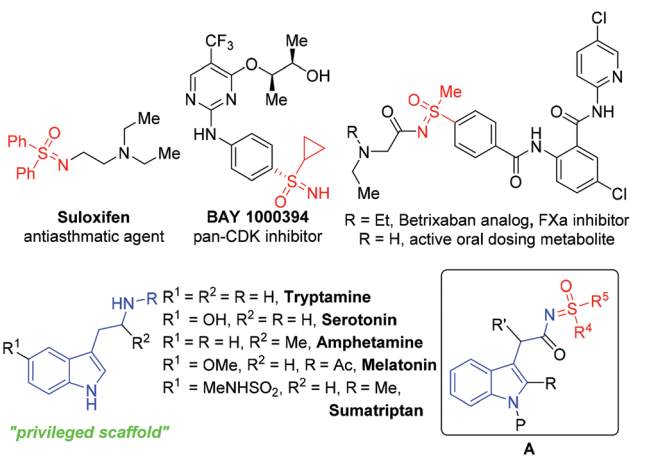


Fig. 1 The active sulfoximine and tryptamine analogues.

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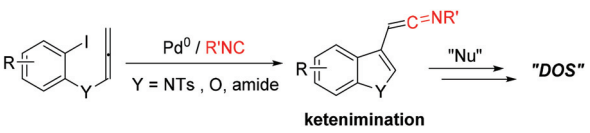
the reaction of *N*H-sulfoximines with carboxylic acids or acid halides.[19] Moreover, oxidative coupling reactions of aldehydes[20] or methyl arenes[21] also offered alternative strategies toward these α-ketoamides. Despite such fruitful approaches being reported, the use of specific types of substrates, such as carbonyl compounds, is generally necessary. Thus, the development of a general synthetic method from readily available precursors is convenient and thus much valuable.

We recently reported a facile synthesis of fused aza-heterocyclic amides *via* a palladium-catalyzed intermolecular cascade reaction of *N*-aryl-*N*-allenamides with isocyanides (Scheme 1).[22] The success of this reaction relies on a direct trapping of an α-indolylpalladium complex by isocyanide, followed by β-H elimination to produce a key ketenimination intermediate.[23] In line with this result and our interest in sulfoximines,[7c,d,24] we wondered whether the α-indolylpalladium complex species could be efficiently trapped by CO (carbon monoxide) as a C1 building block, thus offering an opportunity to construct biologically interesting β-indoly sulfoximidoyl amides through a multicomponent synthetic strategy.[25]

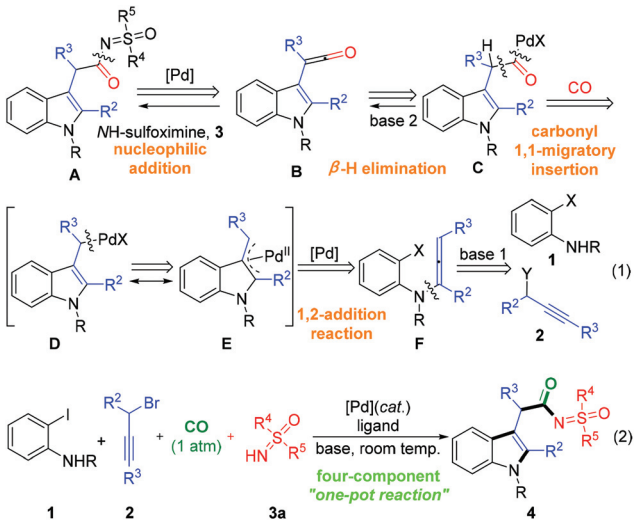
It should be noted that multicomponent reactions (MCRs) have become powerful aza-heterocyclic synthesizing methods in designing new heterocyclic scaffolds and regulating medicinal or biological properties.[26] Compared with conventional step-wise reactions, MCRs can provide a simpler procedure to construct diverse molecules from three or more raw materials in a one-pot procedure, featuring the advantages of simple operation, avoidance of wastage of energy and time, and environmental benignity.

Results and discussion

Our proposed retro-synthetic strategy is presented in Scheme 2, eqn (1). In this approach, a base-assisted coupling reaction of *ortho*-halide aniline **1** with propargyl halide **2** could be expected to form the *ortho*-halide substituted *N*-allenylaniline intermediate **F**, which could undergo a regioselective 1,2-addition reaction to produce the α-indolyl palladium complex **E** and **D**. The species **D** could be reactive enough to be trapped by CO to form ArCOPdX species **C**, which may undergo β-H elimination to deliver species **B**. The following nucleophilic addition reaction of **B** with *N*H-sulfoximine **3** should be able to produce the desired β-indolyl sulfoximidoyl amide **A**. Nevertheless, two challenges must be met: (1) a suitable ligand should be provided to facilitate the trapping of the α-indolylpalladium species with 1 atm of CO under



Scheme 1 Ketenimination reaction of *N*-aryl-*N*-allenamides with isocyanides.



Scheme 2 Proposed synthetic mechanism.

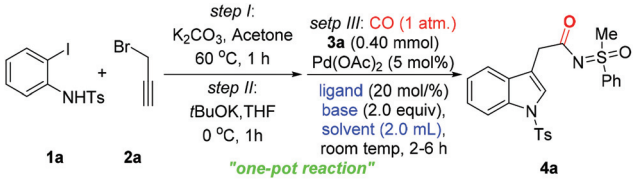
heterogeneous conditions and (2) due to the weak nucleophilic ability of the NH-sulfoximines, a strong base might be capable of boosting β-H elimination to produce the ethenone species, thus favoring the nucleophilic addition step to form the expected C–N bond. Stimulated by the aforementioned challenges, we herein wish to discuss a convenient synthesis of β-indolyl sulfoximidoyl amides from a Pd-catalyzed four-component reaction of 2-iodoaniline **1**, propargyl bromide **2**, carbon monoxide and *N*H-sulfoximine **3** [Scheme 2, eqn (2)]. This multi-step, one-pot cascade reaction proceeds smoothly through the formation of four new chemical bonds in overall good to excellent yields with excellent functional group tolerance under gentle conditions.

To identify the suitable conditions for the proposed Pd-catalyzed sequential domino process, reaction screening involving 2-iodoaniline **1a** and propargyl bromide **2a** as the starting materials was carried out in various bases and solvents. It was found from the initial steps I and II that the intermediate *N*-allenyl-*N*-tosyl-2-iodoaniline (F in Scheme 2) was necessary and crucial for the success of this reaction, because without the formation of F, the desired product **4a** was produced in only low to modest yields during the screening of the reactions. However, with the addition of *t*BuOK, *N*-allenyl-*N*-tosyl-2-iodoaniline could be successfully produced in an almost quantitative yield. Hence, with the perfect generation this intermediate, the next screening of the Pd-catalyzed conditions to efficiently generate the final product **4a** became feasible. With this promising result in hand, we next screened the optimal conditions in step III for further reactions with CO (balloon) and *N*H-sulfoximine **3a** in the presence of $Pd(OAc)_2$, ligand, base and solvent in a one-pot reaction protocol. By utilizing PPh_3 as the ligand, the base was first examined and the results are presented in Table 1. Initial attempts to employ the organic bases pyridine or triethylamine only led to complex results (entries 1 and 2). It was a delight to observe that when K_2CO_3 was used as a base, the desired product **4a** could be iso-

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Table 1 Optimization of the reaction conditions[a]



Entry	[Pd]	Ligand	Base	Solvent	**4a**[a] [%]
1	$Pd(OAc)_2$	PPh_3	Py	Toluene	0
2	$Pd(OAc)_2$	PPh_3	TEA	Toluene	Trace
3	$Pd(OAc)_2$	PPh_3	K_2CO_3	Toluene	27
4	$Pd(CH_3CN)_2Cl_2$	PPh_3	K_2CO_3	Toluene	20
5	$Pd(OCOCF_3)_2$	PPh_3	K_2CO_3	Toluene	19
6	$Pd(dba)_2$	PPh_3	K_2CO_3	Toluene	11
7	$PdCl_2$	PPh_3	K_2CO_3	Toluene	12
8	$Pd(OAc)_2$	PPh_3	Cs_2CO_3	Toluene	34
9	$Pd(OAc)_2$	PPh_3	K_3PO_4	Toluene	29
10	$Pd(OAc)_2$	PPh_3	*t*BuOK	Toluene	40
11	$Pd(OAc)_2$	PPh_3	*t*BuOK	THF	26
12	**$Pd(OAc)_2$**	**PPh_3**	**DBU**	**Toluene**	**94**
13	$Pd(OAc)_2$	PPh_3	DABCO	Toluene	10
14	$Pd(OAc)_2$	$P(4\text{-}MeC_6H_4)_3$	DBU	Toluene	90
15	$Pd(OAc)_2$	$P(4\text{-}MeOC_6H_4)_3$	DBU	Toluene	86
16	$Pd(OAc)_2$	$P(4\text{-}FC_6H_4)_3$	DBU	Toluene	82
17	$Pd(OAc)_2$	PCy_3	DBU	Toluene	Trace
18	$Pd(OAc)_2$	DPPM	DBU	Toluene	84
19	$Pd(OAc)_2$	SPhos	DBU	Toluene	75
20	$Pd(OAc)_2$	PPh_3	DBU	THF	80
21	$Pd(OAc)_2$	PPh_3	DBU	1,4-Dioxane	87
22	$Pd(OAc)_2$	PPh_3	DBU	DMF	82
23	$Pd(OAc)_2$	PPh_3	DBU	CH_3CN	92
24	$Pd(OAc)_2$	PPh_3	DBU	THF : toluene = 1 : 2	88[b]
25	$Pd(OAc)_2$	PPh_3	DBU	Toluene	66[c]

[a] Reaction conditions: step I: 2-iodoaniline **1a** (0.44 mmol), propargyl bromide **2a** (0.44 mmol), K_2CO_3 (0.44 mmol), acetone (1.0 mL), 60 °C, 2 h; step II: *t*BuOK (0.56 mmol), THF (1.5 mL), 0 °C to r.t., 2 h; step III: CO balloon, *N*H-sulfoximine **3a** (0.40 mmol), $Pd(OAc)_2$ (5 mol%), ligand (20 mol%), base (0.80 mmol), solvent (2.0 mL), r.t., 2–3 h. [b] THF (1.0 mL) in step II was not removed while toluene (1.0 mL) was added. [c] $Mn(CO)_5Br$ (1.0 equiv.) was utilized instead of CO. DBU = 1,8-diazabicyclo[5.4.0]undec-7-ene; DPPM = bis(diphenylphosphino)methane.

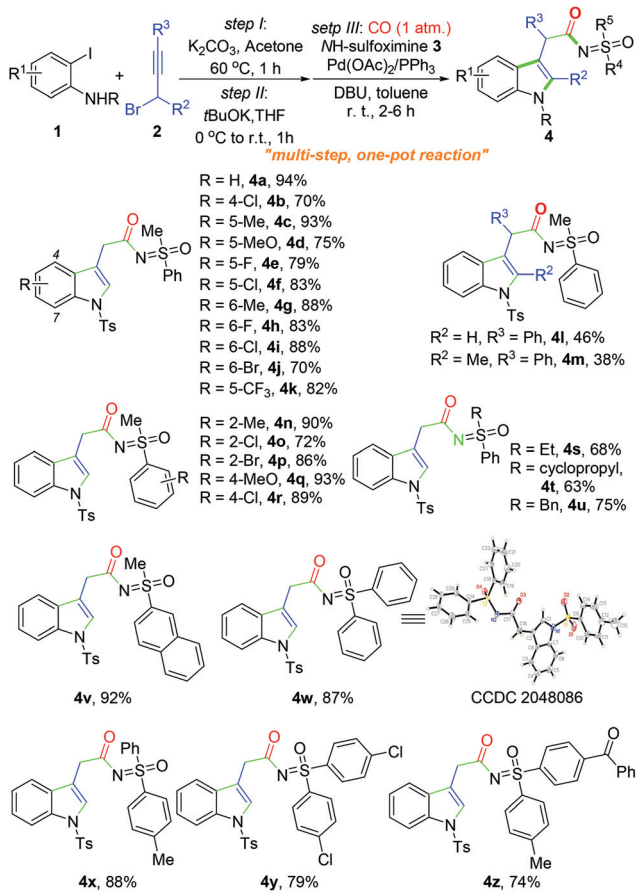
lated in 27% yield (entry 3). Based on this experiment, we subsequently screened various Pd catalysts, which again demonstrated that $Pd(OAc)_2$ was an optimal Pd catalyst (entries 4–7). Other inorganic bases, such as Cs_2CO_3 and K_3PO_4, were found to be workable for this reaction to produce the desired product **4a**, albeit in poor yields (entries 8 and 9).

With *t*BuOK as the base in toluene, the product **4a** was delivered in 40% yield, whilst an inferior result was observed when THF, which was employed as the solvent in the previous step II, was employed in place of toluene (entries 10 and 11). To our delight, a strong organic base, DBU, was found to be the optimal base for this reaction. A smooth conversion was observed and the product **4a** was isolated in 94% yield when DBU was added (entry 12). Another strong organic base DABCO was not effective (entry 13); hence, we used DBU for further optimizations.

Encouraged by these results, we further examined the phosphine ligand. The results indicated that electron-rich P ligands were all effective for the reaction, whereas PPh_3 again proved to be the optimal choice (entries 14–19). For the solvent effect in this reaction, carrying out the reaction in THF, 1,4-dioxane, and DMF did not facilitate the formation of **4a**, respectively (entries 20–22). An identical 92% yield of **4a** was obtained with CH_3CN as the solvent (entry 23). Finally, we tried the mixture solvent of THF : toluene = 1 : 2; however, no satisfactory result was observed (entry 24). Instead of using the toxic gas CO, we used $Mn(CO)_5Br$ (1.0 equiv.) as a carbonyl source under N_2, and the desired product **4a** could be isolated in 66% yield (entry 25).

With the optimized conditions in hand (Table 1, entry 12), we next studied the scope of this Pd-catalyzed multicomponent cascade annulation reaction (Scheme 3). The Pd-catalyzed one-pot reactions of various N-Ts protected 2-iodoanilines **1** with propargyl bromide **2a**, *N*H-sulfoximine **3a** and 1 atm of CO were firstly evaluated. As shown in Scheme 2, a broad range of β-indolyl sulfoximidoyl amides **4** could be readily obtained. The *ortho*-iodoanilines **1** with electron-donating groups at the *para*- (**4c** and **4d**) or *meta*-positions (**4g**) of the phenyl moiety all delivered the corresponding products in good yields. The electron-withdrawing halide substituents, such as F (**4e**, **4h** and **4k**) and Cl (**4b**, **4f**, and **4i**), on the phenyl moiety could afford the corresponding products in good to high yields.

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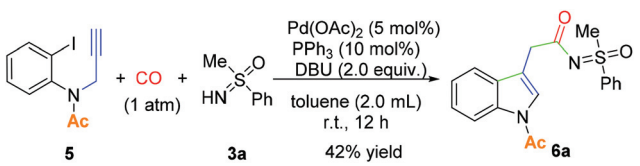
Scheme 3 Substrate scope studies. Reaction conditions: Table 1, entry 8, step I: 2-iodoaniline **1** (0.44 mmol), propargyl bromide **2** (0.44 mmol), K_2CO_3 (0.44 mmol), acetone (1.0 mL), 60 °C, 2 h; step II: *t*BuOK (0.56 mmol), THF (1.5 mL), 0 °C to r.t., 2 h; step III: CO (1 atm), *N*H-sulfoximine **3** (0.40 mmol), $Pd(OAc)_2$ (5 mol%), PPh_3 (20 mol%), DBU (0.80 mmol), toluene (2.0 mL), r.t., 2–6 h.

Gratifyingly, the synthetically useful bromide atom could also be tolerated under these Pd-catalyzed conditions, delivering the desired products **4j** and **4p** in 70% and 86% yields, respectively. In addition, non-terminal propargyl bromides **2** turned out to be suitable starting materials as well; R^2 and R^3 could be a methyl or phenyl group, thus leading to poly-substituted β-indolyl sulfoximidoyl amides **4l** and **4m** in acceptable yields.

The variations on the *S*-linkage of *N*H-sulfoximines **3** were further studied under the Pd-catalyzed one-pot reaction conditions. For the *S*-methyl substituted aryl *N*H-sulfoximines, various substitutions on the benzene ring at different positions with varied electronic natures were readily allowed, leading to the desired products **4n–4r** with good to excellent efficiency. The halide substituents attached at the phenyl ring could be Cl (**4o**, **4r**) or Br (**4p**), showing their further synthetic potential through Pd-catalyzed cross-coupling reactions with organometallic agents. Except for the *S*-methyl group in substrate **3a**, other alkyl groups that were linked at the S atom in substrates **3** were also divergent, including ethyl (**4s**), sterically hindered cyclopropyl (**4t**) and benzyl (**4u**) groups. *N*H-Sulfoximine **3** that contains an *S*-naphthyl group underwent the reaction smoothly to give product **4v** in 92% yield. Diphenyl *N*H-sulfoximine was a viable substrate, leading to the corresponding products **4w** and **4y** in good yields, respectively. The structure of compound **4w** was established unambiguously by X-ray crystal diffraction analysis.[27] In addition to the results, the benzoylphenone derivative **4z** (74% yield) showed the compatible property of ketone substitutions.

Having demonstrated a broad reaction scope obtained using Ts-protected *ortho*-iodoaniline **1** as the key substrate, our attention turned to the easily removable protecting groups which were linked at the indole N moieties. It was found that this reaction did not work with an acyl-protected compound, *N*-(2-iodophenyl)acetamide, with propargyl bromide **2a**, CO (1 atm) and *N*H-sulfoximine **3a** under the Pd-catalyzed conditions. Meanwhile, only a moderate yield of acetyl protected indole **6a** was obtained in the Pd-catalyzed three-component tandem annulation reaction of *N*-(2-iodophenyl)-*N*-(prop-2-yn-1-yl)acetamide **5** with 1 atm of CO and *N*H-sulfoximine **3a** (Scheme 4). However, the situation could be significantly changed when *N*-aryl-*N*-allenamide **7** was employed as a useful substrate with 1 atm of CO and *N*H-sulfoximine **3a** under the Pd-catalyzed conditions (Scheme 4). For this approach, the starting materials acyl-protected *N*-allenamide **7** and **3a** were mixed in toluene at room temperature; then $Pd(OAc)_2/PPh_3$ and DBU were added under a CO balloon atmosphere. To our delight, the Ac-protected indole **6a** was successfully formed in 84% yield. The reaction of Boc-protected *N*-allenamide **7** and **3a** with CO was also compatible under the Pd-catalyzed conditions, affording the desired Boc-protected indoles **6b–6e** in good to excellent yields. Unfortunately, for the H- or alkyl-substituted substrates **7**, no desired product **6f** was isolated (Scheme 5).

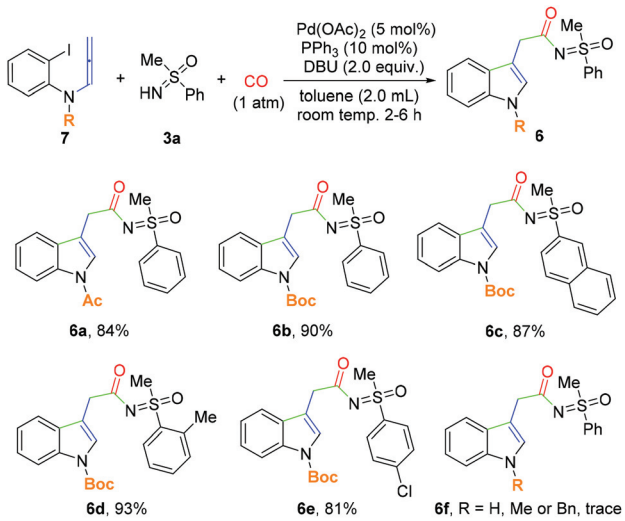
A gram-scale reaction of Ts-protected *N*-allenamide **7a** with *N*H-sulfoximine **3a** under a CO balloon atmosphere afforded the desired β-indolyl acetyl amide **4a** in 86% yield (Scheme 6, eqn (1)). The products prepared by this procedure are valuable precursors for the synthesis of biologically interesting sulfoximine analogues that contain a tryptamine moiety. For instance, compound **4a** could be readily reduced by $BH_3{\cdot}Me_2S$ in dichloromethane to afford a tryptamine derivate **8** in 65% yield (Scheme 6, eqn (2)). Considering the importance of the tryptamine and sulfoximine structural motifs as pharmaceutical candidates (Fig. 1), we hope that the resulting product **8** should be able to exhibit potential biological or medicinal activities in future medicinal studies.



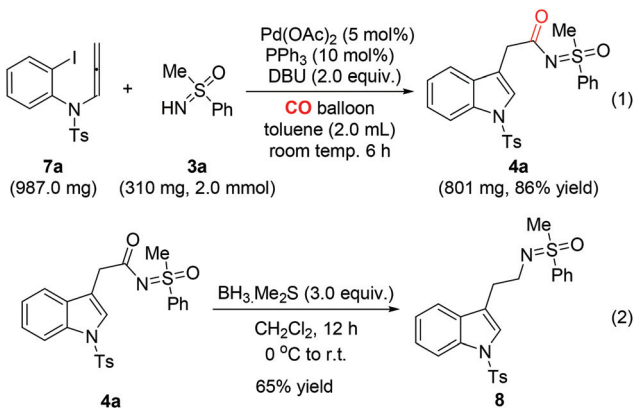
Scheme 4 The reactivity of Ac-protected propyl aniline with **3a** and CO.

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Scheme 5 Variation of the protecting groups on indole N atoms. Reaction conditions: *N*-allenylaniline **7** (0.48 mmol), CO (1 atm), *N*H-sulfoximine **3a** (0.40 mmol), $Pd(OAc)_2$ (5 mol%), PPh_3 (20 mol%), DBU (0.80 mmol), toluene (2.0 mL), r.t., 2–6 h.



Scheme 6 Gram-scale synthesis of **4a** (eqn (1)) and its further transformation to the tryptamine-containing sulfoximine analogue **8** (eqn (2)).

Conclusions

In summary, we have described a room temperature Pd-catalyzed multi-step one-pot cascade reaction of *ortho*-iodoaniline, propargyl bromide, 1 atm of CO and *N*H-sulfoximine, which enables operational convenience toward β-indolyl sulfoximidoyl amides in mostly good to excellent yields under carefully modified reaction conditions. The success of this reaction was attributed to the intermolecular trapping of an α-indolylpalladium complex by CO. Because the obtained products could be potential biologically active sulfoximine derivatives that contain a tryptamine moiety, the variability of the starting materials and their easy accessibility make this protocol extremely attractive. Protecting groups such as Ac and Boc located at the indole N atom could be compatible for the reactions of *N*-aryl-*N*-allenamide with CO and *N*H-sulfoximine. Considering the easy availability of the starting materials and the gentle but highly efficient catalytic systems, this one-pot synthetic protocol could serve as a valuable and powerful method for the expedient synthesis of biologically active compounds.

Experimental section

General procedure for the synthesis of β-indolyl sulfoximidoyl amide 4a

Step I: To a stirred solution of 2-iodoaniline[28] **1a** (164 mg, 0.44 mmol) and K_2CO_3 (60.72 mg, 0.44 mmol) in 1.0 mL of acetone was added 3-bromoprop-1-yne **2a** (45 μl, 1.2 mmol) at room temperature under a N_2 atmosphere. The resulting solution was stirred at 60 °C for 1 hour. Upon completion of this reaction as monitored by TLC, acetone was removed using a rotary evaporator.

Step II: Following the first step I, anhydrous THF (1.0 mL) was added into the flask under a nitrogen atmosphere; the solution was cooled to 0 °C, and then *t*BuOK in THF (0.5 mL) was added dropwise (64.6 mg, 0.56 mmol). The reaction mixture was stirred, allowed to warm to room temperature and stirred for an additional 1 hour. A light yellow solution was obtained in this step. Upon the disappearance of the key intermediate as monitored *via* TLC, the solvent THF was removed using a rotary evaporator to give a crude residue, which was used in the next step without purification.

Step III: Following step II, $Pd(OAc)_2$ (4.48 mg, 5 mol%), PPh_3 (20.96 mg, 20 mol%), DBU (121.8 mg, 0.80 mmol), *N*H-sulfoximine **3a** (62.0 mg, 0.40 mmol), and dry toluene (2.0 mL) were added sequentially. The resulting brown solution was purged with a CO balloon 3 times, and then stirred at room temperature for 2–3 h. A yellow solution was obtained. Upon completion of the reaction as indicated by TLC, the reaction mixture was filtered through a short pad of silica gel and washed with ethyl acetate (15.0 mL) unless otherwise noted. The concentration of the filtrate under reduced pressure provided a crude product, which was purified by silica gel column chromatography (petroleum ether/ethyl acetate = 10 : 1–3 : 1, v/v) to afford the desired compound **4a** as a white solid.

The other products **4b** to **4y** were prepared through a similar procedure to that of product **4a**.

***N*-(Methyl(oxo)(phenyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4a).** Yield: 175.3 mg, 94%; white solid, mp: 52.6–54.6 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.97 (d, J = 8.3 Hz, 1H), 7.75 (t, J = 7.1 Hz, 4H), 7.63–7.57 (m, 2H), 7.54 (d, J = 7.8 Hz, 1H), 7.47 (t, J = 7.7 Hz, 2H), 7.30 (t, J = 7.7 Hz, 1H), 7.22–7.15 (m, 3H), 3.75 (s, 2H), 3.27 (s, 3H), 2.29 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.2, 144.8, 138.4, 135.4, 135.1, 133.8, 131.0, 129.8, 129.6, 127.0, 126.8, 124.6, 124.6, 123.1, 120.1, 117.1, 113.6, 44.1, 36.3, 21.5. IR (KBr): ν 3055.6, 2924.0, 1634.7, 1446.7, 1359.3, 1177.3, 1083.7, 774.4, 571.2, 485.1 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{23}N_2O_4S_2$, 467.1094; found, 467.1102.

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2-(4-Chloro-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4b). Yield: 140.2 mg, 70%; white solid, mp: 69–71 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.79 (d, J = 7.6 Hz, 1H), 7.67 (d, J = 7.9 Hz, 2H), 7.60 (d, J = 7.3 Hz, 2H), 7.51–7.45 (m, 2H), 7.35 (t, J = 7.3 Hz, 2H), 7.10–7.02 (m, 4H), 4.00–3.89 (m, 2H), 3.26 (s, 3H), 2.16 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.6, 145.1, 138.7, 136.5, 135.0, 133.7, 129.9, 129.5, 127.9, 127.1, 126.8, 126.4, 125.1, 124.2, 117.0, 112.4, 43.9, 37.1, 21.5. IR (KBr): ν 3062.4, 2925.9, 1644.5, 1447.2, 1371.0, 1220.5, 1097.9, 997.1, 746.2, 575.1 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}ClN_2O_4S_2$, 501.0704; found, 501.0710.

***N*-(Methyl(oxo)(phenyl)-l6-sulfanylidene)-2-(5-methyl-1-tosyl-1*H*-indol-3-yl)acetamide (4c).** Yield: 178.6 mg, 93%; white solid, mp: 62–64 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.84 (d, J = 8.4 Hz, 1H), 7.77–7.74 (m, 2H), 7.70 (d, J = 8.4 Hz, 2H), 7.58 (t, J = 7.5 Hz, 1H), 7.52 (s, 1H), 7.44 (t, J = 7.8 Hz, 2H), 7.30 (s, 1H), 7.13–7.09 (m, 3H), 3.72 (s, 2H), 3.25 (s, 3H), 2.35 (s, 3H), 2.24 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.2, 144.7, 138.5, 135.4, 133.8, 133.4, 132.7, 131.3, 129.8, 129.6, 127.0, 126.7, 126.0, 124.7, 120.0, 117.1, 113.3, 44.0, 36.3, 21.5, 21.3. IR (KBr): ν 3023.7, 2924.3, 1643.0, 1447.6, 1307.6, 1120.4, 977.7, 811.7, 685.2, 509.4 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_4S_2$, 481.1250; found, 481.1254.

2-(5-Methoxy-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4d). Yield: 148.8 mg, 75%; white solid, mp: 61–63 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.86 (d, J = 9.0 Hz, 1H), 7.75–7.69 (m, 4H), 7.60 (t, J = 7.5 Hz, 1H), 7.52 (s, 1H), 7.45 (t, J = 7.9 Hz, 2H), 7.14 (d, J = 8.1 Hz, 2H), 6.96 (d, J = 2.4 Hz, 1H), 6.91–6.89 (m, 1H), 3.72 (s, 3H), 3.70 (s, 2H), 3.26 (s, 3H), 2.28 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.1, 156.4, 144.7, 138.4, 135.3, 133.8, 132.0, 129.8, 129.8, 129.6, 127.0, 126.7, 125.4, 117.3, 114.5, 113.8, 102.5, 55.6, 44.1, 36.4, 21.5. IR (KBr): ν 3011.9, 2928.1, 1643.0, 1447.9, 1308.6, 1118.5, 977.8, 813.8, 669.2, 511.2 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_5S_2$, 497.1199; found, 497.1201.

2-(5-Fluoro-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4e). Yield: 153.0 mg, 79%; white solid, mp: 54–56 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.92–7.88 (m, 1H), 7.80–7.78 (m, 2H), 7.71 (d, J = 8.3 Hz, 2H), 7.63–7.60 (m, 2H), 7.48 (t, J = 7.8 Hz, 2H), 7.20–7.15 (m, 3H), 7.03–6.98 (m, 1H), 3.70 (s, 2H), 3.28 (s, 3H), 2.28 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.8, 159.6 (d, ^1J = 238 Hz), 145.0, 138.4, 135.1, 133.9, 131.4, 129.9, 129.6, 127.0, 126.8, 126.4, 117.0, 114.7, 114.6, 112.6 (d, ^2J = 26 Hz), 105.9 (d, ^2J = 24 Hz), 44.1, 36.1, 21.5. ^{19}F NMR (376 MHz, $CDCl_3$) δ −119.6 (s). IR (KBr): ν 2929.4.4, 1653.8, 1444.9, 1369.2, 1225.7, 1108.7, 1015.7, 895.6, 753.5, 590.0 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}FN_2O_4S_2$, 485.1000; found, 485.1007.

2-(5-Chloro-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4f). Yield: 166.0 mg, 83%; white solid, mp: 61–63 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.88 (d, J = 8.8 Hz, 1H), 7.80–7.78 (m, 2H), 7.71 (d, J = 8.3 Hz, 2H), 7.63–7.58 (m, 2H), 7.52–7.47 (m, 3H), 7.25–7.22 (m, 1H), 7.16 (d, J = 7.9 Hz, 2H), 3.71 (s, 2H), 3.27 (s, 3H), 2.28 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.7, 145.2, 138.3, 135.1, 133.9, 133.5, 132.3, 130.0, 129.7, 129.0, 127.0, 126.8, 126.0, 124.8, 119.9, 116.6, 114.7, 44.1, 36.1, 21.5. IR (KBr): ν 3069.1, 2928.8, 1650.0, 1445.9, 1369.4, 1266.8, 1100.2, 994.8, 734.1, 538.0 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}ClN_2O_4S_2$, 501.0710; found, 501.0714.

***N*-(Methyl(oxo)(phenyl)-l6-sulfanylidene)-2-(6-methyl-1-tosyl-1*H*-indol-3-yl)acetamide (4g).** Yield: 169.0 mg, 88%; white solid, mp: 66–68 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.78–7.71 (m, 5H), 7.59 (t, J = 7.5 Hz, 1H), 7.49–7.39 (m, 4H), 7.14 (d, J = 8.2 Hz, 2H), 7.02 (d, J = 8.0 Hz, 1H), 3.72 (s, 2H), 3.25 (s, 3H), 2.45 (s, 3H), 2.26 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.2, 144.7, 138.5, 135.5, 134.7, 133.8, 129.8, 129.6, 128.8, 127.0, 126.7, 124.7, 123.9, 119.8, 117.1, 113.7, 44.0, 36.3, 21.9, 21.5. IR (KBr): ν 3062.6, 2924.4, 1644.0, 1447.4, 1364.4, 1171.7, 996.3, 812.4, 668.6, 539.7 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_4S_2$, 481.1250; found, 481.1253.

2-(6-Fluoro-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4h). Yield: 160.7 mg, 83%; white solid, mp: 59–61 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.78 (d, J = 7.5 Hz, 2H), 7.74 (d, J = 8.3 Hz, 2H), 7.71–7.68 (m, 1H), 7.64–7.60 (m, 1H), 7.54 (s, 1H), 7.51–7.44 (m, 3H), 7.19 (d, J = 8.2 Hz, 2H), 6.97–6.92 (m, 1H), 3.73 (s, 2H), 3.28 (s, 3H), 2.30 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.0, 160.9 (d, ^1J = 240 Hz), 145.1, 138.3, 135.1, 133.9, 130.0, 129.6, 127.0, 126.8, 124.8, 124.7, 121.1, 121.0, 116.9, 111.5 (d, ^2J = 24 Hz), 100.9 (d, ^2J = 24 Hz), 44.1, 36.2, 21.6. ^{19}F NMR (376 MHz, $CDCl_3$) δ −116.5 (s). IR (KBr): ν 3063.6, 2925.9, 1644.2, 1484.2, 1365.7, 1222.4, 1103.8, 894.3, 738.7, 584.1 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}FN_2O_4S_2$, 485.1000; found, 485.1006.

2-(6-Chloro-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4i). Yield: 176 mg, 88%; white solid, mp: 66–68 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.15 (d, J = 1.4 Hz, 1H), 7.79–7.76 (m, 2H), 7.73 (d, J = 8.3 Hz, 2H), 7.64–7.61 (m, 1H), 7.54 (s, 1H), 7.51–7.47 (m, 2H), 7.39 (d, J = 8.4 Hz, 1H), 7.31–7.28 (m, 1H), 7.19 (d, J = 8.3 Hz, 2H), 3.72 (s, 2H), 3.27 (s, 3H), 2.31 (s, 3H).^{13}C NMR (100 MHz, $CDCl_3$) δ 178.9, 145.2, 138.3, 135.4, 135.1, 133.9, 130.6, 130.0, 129.6, 129.5, 127.0, 126.8, 125.1, 123.8, 121.1, 116.9, 113.7, 44.1, 36.1, 21.5. IR (KBr): ν 3064.1, 2927.9, 1644.8, 1447.9, 1370.7, 1173.0, 979.9, 814.2, 685.6, 585.0 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}ClN_2O_4S_2$, 501.0704; found, 501.0707.

2-(6-Bromo-1-tosyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (4j). Yield: 152.3 mg, 70%; white solid, mp: 66–68 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.14 (d, J = 1.5 Hz, 1H), 7.79–7.76 (m, 2H), 7.73 (d, J = 8.4 Hz, 2H), 7.64–7.60 (m, 1H), 7.54 (s, 1H), 7.48 (t, J = 7.9 Hz, 2H), 7.39 (d, J = 8.4 Hz, 1H), 7.30–7.28 (m, 1H), 7.18 (d, J = 8.1 Hz, 2H), 3.72 (s, 2H), 3.27 (s, 3H), 2.30 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.8, 145.2,

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138.4, 135.7, 135.1, 133.9, 130.0, 129.9, 129.6, 127.0, 126.8, 126.4, 125.1, 121.4, 118.3, 117.0, 116.6, 44.1, 36.1, 21.6. IR (KBr): ν 3063.9, 2927.3, 1644.9, 1448.5, 1371.4, 1172.2, 996.6, 813.7, 685.3, 578.7 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{22}BrN_2O_4S_2$, 545.0199; found, 545.0202.

***N*-(Methyl(oxo)(phenyl)-l6-sulfanylidene)-2-(1-tosyl-5-(trifluoromethyl)-1*H*-indol-3-yl)acetamide (4k).** Yield: 175.2 mg, 82%; white solid; mp: 65–67 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2. ^1H NMR (400 MHz, $CDCl_3$) δ 8.07 (d, J = 8.7 Hz, 1H), 7.87 (s, 1H), 7.80–7.75 (m, 4H), 7.68 (s, 1H), 7.62 (t, J = 7.4 Hz, 1H), 7.55–7.47 (m, 3H), 7.20 (d, J = 8.1 Hz, 2H), 3.79 (s, 2H), 3.28 (s, 3H), 2.31 (s, 3H).^{13}C NMR (100 MHz, $CDCl_3$) δ 178.7 (s), 145.4 (s), 138.2 (s), 136.5 (s), 135.0 (s), 133.9 (s), 130.7 (s), 130.1 (s), 129.7 (s), 126.9 (s), 126.9 (s), 126.3 (s), 125.4 (d, ^1J = 32 Hz), 124.4 (d, ^2J = 70 Hz), 121.3 (d, ^2J = 4 Hz), 117.8 (d, ^2J = 4 Hz), 117.1 (s), 113.9 (s), 44.1 (s), 36.0 (s), 21. 6 (s).^{19}F NMR (376 MHz, $CDCl_3$) δ −60.9 (s). IR (KBr): ν 3065.4, 2928.8, 1645.3, 1448.2, 1313.4, 1173.3, 1097.4, 978.5, 816.1, 686.1 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{25}H_{21}F_3N_2O_4S_2$, 535.0968; found, 535.0971.

***N*-(Methyl(oxo)(phenyl)-l6-sulfanylidene)-2-phenyl-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4l).** Yield: 99.7 mg, 46%; white solid, mp: 64–66 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 7.96–7.93 (m, 1H), 7.76 (s, 1H), 7.72 (d, J = 8.3 Hz, 2H), 7.61 (d, J = 8.5 Hz, 2H), 7.52–7.47 (m, 1H), 7.45–7.41 (m, 1H), 7.38–7.36 (m, 2H), 7.33–7.22 (m, 6H), 7.17 (d, J = 8.1 Hz, 2H), 7.13–7.09 (m, 1H), 5.17 (s, 1H), 3.24 (d, J = 18.6 Hz, 3H), 2.31 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.8, 144.7, 138.4, 135.4, 133.8, 129.8, 129.6, 129.6, 128.8, 128.7, 128.5, 127.2, 127.0, 127.0, 126.8, 125.1, 124.6, 123.1, 121.2, 120.0, 113.7, 52.8, 44.3, 21.5. IR (KBr): ν 2927.5, 1645.5, 1446.3, 1369.1, 1221.4, 1093.4, 895.8, 734.0, 668.9, 538.0 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{30}H_{27}N_2O_4S_2$, 543.1407; found, 543.1412.

***N*-(Methyl(oxo)(phenyl)-l6-sulfanylidene)-2-(2-methyl-1-tosyl-1*H*-indol-3-yl)-2-phenylacetamide (4m).** Yield: 84.5 mg, 38%; white solid, mp: 55–57 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 8.23 (d, J = 8.4 Hz, 1H), 7.61–7.55 (m, 3H), 7.50 (d, J = 7.4 Hz, 2H), 7.43–7.36 (m, 3H), 7.25–7.20 (m, 4H), 7.17–7.09 (m, 5H), 5.28 (s, 1H), 3.22 (s, 3H), 2.57 (s, 3H), 2.23(s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 180.3, 144.6, 138.5, 138.0, 136.7, 136.3, 135.2, 133.7, 130.0, 129.8, 129.4, 128.7, 128.1, 127.0, 126.7, 126.3, 123.9, 123.3, 120.5, 119.1, 114.7, 52.0, 43.60, 21.6, 13.3. IR (KBr): ν 2928.6, 1647.5, 1449.3, 1379.1, 1226.4, 1095.3, 897.8, 738.5, 669.9, 533.0 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{31}H_{29}N_2O_4S_2$, 557.1563; found, 557.1569.

***N*-(Methyl(oxo)(*o*-tolyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4n).** Yield: 172.8 mg, 90%; white solid, mp: 65–67 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 7.95 (d, J = 8.3 Hz, 1H), 7.90 (d, J = 8.0 Hz, 1H), 7.72 (d, J = 8.3 Hz, 2H), 7.55 (s, 1H), 7.51 (d, J = 7.8 Hz, 1H), 7.46–7.43 (m, 1H), 7.29–7.25 (m, 2H), 7.22–7.17 (m, 2H), 7.13 (d, J = 8.2 Hz, 2H), 3.73 (s, 2H), 3.27 (s, 3H), 2.37 (s, 3H), 2.26 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.8, 144.8, 137.0, 136.4, 135.4, 135.1, 133.7, 133.2, 131.0, 129.8, 129.0, 126.9, 126.8, 124.6, 124.6, 123.1, 120.1, 117.1, 113.5, 42.9, 36.2, 21.5, 20.0. IR (KBr): ν 3061.0, 2927.8, 1643.9, 1447.7, 1366.0, 1173.8, 1096.0, 976.4, 758.8, 577.3 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_4S_2$, 481.1250; found, 481.1259.

***N*-((2-Chlorophenyl)(methyl)(oxo)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4o).** Yield: 144.0 mg, 72%; white solid, mp: 63–65 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 8.14–8.12 (m, 1H), 7.93 (d, J = 8.3 Hz, 1H), 7.72 (d, J = 8.2 Hz, 2H), 7.55 (s, 1H), 7.53–7.48 (m, 2H), 7.45–7.40 (m, 2H), 7.28–7.24 (m, 1H), 7.19–7.13 (m, 3H), 3.73 (s, 2H), 3.39 (s, 3H), 2.27 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.4, 144.7, 135.9, 135.5, 135.1, 134.8, 132.1, 131.6, 131.1, 131.1, 129.8, 127.9, 126.8, 124.7, 124.5, 123.1, 120.0, 116.8, 113.5, 42.1, 35.7, 21.5. IR (KBr): ν 3063.8, 2928.6, 1646.5, 1449.3, 1366.0, 1173.6, 1044.2, 976.1, 759.6, 577.7 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{22}ClN_2O_4S_2$, 501.0704; found, 501.0709.

***N*-((2-Bromophenyl)(methyl)(oxo)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4p).** Yield: 187.1 mg, 86%; white solid, mp: 66–68 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 8.18 (d, J = 7.9 Hz, 1H), 7.94 (d, J = 8.3 Hz, 1H), 7.73 (d, J = 8.2 Hz, 2H), 7.67 (d, J = 7.8 Hz, 1H), 7.56 (s, 1H), 7.52–7.47 (m, 2H), 7.45–7.41 (m, 1H), 7.28–7.26 (m, 1H), 7.19–7.15 (m, 3H), 3.74 (s, 2H), 3.42 (s, 3H), 2.29 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.3, 144.7, 137.6, 135.7, 135.5, 135.0, 134.7, 131.8, 131.1, 129.8, 128.5, 126.8, 124.7, 124.5, 123.0, 120.0, 119.2, 116.7, 113.5, 41.7, 35.7, 21.5. IR (KBr): ν 3062.2, 2924.8, 1644.5, 1493.2, 1364.7, 1171.5, 1094.7, 972.8, 757.2, 572.1 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{22}BrN_2O_4S_2$, 545.0199; found, 545.0206.

***N*-((4-Methoxyphenyl)(methyl)(oxo)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4q).** Yield: 184.5 mg, 93%; white solid, mp: 55–57 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 7.97 (d, J = 8.2 Hz, 1H), 7.73 (d, J = 7.5 Hz, 2H), 7.67–7.65 (m, 2H), 7.57–7.53 (m, 2H), 7.30 (d, J = 7.4 Hz, 1H), 7.22–7.14 (m, 3H), 6.90 (d, J = 7.9 Hz, 2H), 3.83 (d, J = 0.9 Hz, 3H), 3.74 (s, 2H), 3.25 (d, J = 0.8 Hz, 3H), 2.28 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.1, 163.9, 144.8, 135.4, 135.1, 131.0, 129.8, 129.4, 129.2, 126.8, 124.6, 123.1, 120.1, 117.3, 114.8, 113.6, 55.8, 44.4, 36.3, 21.5. IR (KBr): ν 3055.1, 2927.9, 1643.4, 1447.0, 1365.6, 1173.3, 1097.4, 976.8, 671.7, 538.5 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_5S_2$, 497.1199; found, 497.1208.

***N*-((4-Chlorophenyl)(methyl)(oxo)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4r).** Yield: 178.0 mg, 89%; white solid, mp: 59–61 °C. Eluent : ethyl acetate/petroleum ether = 1 : 2 ^1H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, J = 8.2 Hz, 1H), 7.75 (d, J = 7.4 Hz, 2H), 7.63 (d, J = 7.5 Hz, 2H), 7.56 (s, 1H), 7.52 (d, J = 7.6 Hz, 1H), 7.39 (d, J = 7.5 Hz, 2H), 7.33–7.29 (m, 1H), 7.22–7.17 (m, 3H), 3.74 (s, 2H), 3.25 (s, 3H), 2.30 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.1, 144.9, 140.6, 136.9, 135.3, 135.1, 130.9, 129.8, 128.6, 126.8, 124.7, 124.6, 123.1, 120.1, 117.0, 113.6, 44.0, 36.2, 21.5. IR (KBr): ν 2927.7, 1645.1, 1448.0, 1366.4, 1173.2, 1094.8, 977.5, 814.7, 670.8, 538.2 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + H]^+$ calcd for $C_{24}H_{22}ClN_2O_4S_2$, 501.0704; found, 501.0712.

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***N*-(Ethyl(oxo)(phenyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4s).** Yield: 130.6 mg, 68%; white solid, mp: 40–42 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.98 (d, J = 8.3 Hz, 1H), 7.74 (d, J = 8.3 Hz, 2H), 7.69 (d, J = 7.4 Hz, 2H), 7.61–7.54 (m, 3H), 7.46–7.43 (m, 2H), 7.32–7.28 (m, 1H), 7.22–7.18 (m, 1H), 7.15 (d, J = 8.1 Hz, 2H), 3.80–3.71 (m, 2H), 3.45–3.33 (m, 2H), 2.27 (s, 3H), 1.13–1.09 (m, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.1, 144.8, 136.0, 135.4, 135.1, 133.7, 131.0, 129.8, 129.5, 127.8, 126.8, 124.6, 124.5, 123.1, 120.2, 117.4, 113.6, 50.3, 36.3, 21.5, 7.0. IR (KBr): ν 3062.9, 2936.6, 1644.1, 1447.3, 1365.8, 1173.3, 1095.7, 975.9, 744.7, 577.1 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_4S_2$, 481.1256; found, 481.1253.

***N*-(Cyclopropyl(oxo)(phenyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4t).** Yield: 124.0 mg, 63%; white solid, mp: 58–60 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.97 (d, J = 8.3 Hz, 1H), 7.74 (d, J = 8.3 Hz, 2H), 7.68–7.66 (m, 2H), 7.57–7.50 (m, 3H), 7.41 (t, J = 7.8 Hz, 2H), 7.31–7.27 (m, 1H), 7.20–7.14 (m, 3H), 3.77–3.65 (m, 2H), 2.55–2.50 (m, 1H), 2.27 (s, 3H), 1.40–1.35 (m, 1H), 1.20–1.05 (m, 2H), 0.97–0.82 (m, 2H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.5, 144.8, 138.7, 135.4, 135.11, 133.4, 131.0, 129.8, 129.4, 127.1, 126.8, 124.6, 124.5, 123.1, 120.1, 117.3, 113.5, 36.3, 33.1, 21.5, 6.8, 5.2. IR (KBr): ν 3060.6, 2920.6, 1645.7, 1446.9, 1365.9, 1220.8, 1095.0, 975.6, 745.8, 577.1 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{26}H_{25}N_2O_4S_2$, 493.1250; found, 493.1257.

***N*-(Benzyl(oxo)(phenyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4u).** Yield: 162.6 mg, 75%; white solid, mp: 59–61 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.00 (d, J = 8.2 Hz, 1H), 7.73 (d, J = 8.1 Hz, 2H), 7.59 (s, 1H), 7.52 (d, J = 4.6 Hz, 2H), 7.38 (d, J = 7.4 Hz, 2H), 7.30–7.27 (m, 3H), 7.22–7.16 (m, 2H), 7.11–7.01 (m, 4H), 6.76 (d, J = 7.3 Hz, 2H), 4.76 (d, J = 13.5 Hz, 1H), 4.60 (d, J = 13.6 Hz, 1H), 3.77 (s, 2H), 2.21 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.5, 144.8, 135.4, 135.2, 135.1, 133.8, 131.1, 131.0, 129.8, 129.1, 129.0, 128.4, 128.4, 127.0, 126.8, 124.6, 124.6, 123.1, 120.2, 117.4, 113.6, 61.9, 36.5, 21.5. IR (KBr): ν 3063.5, 2927.6, 1642.8, 1447.5, 1365.6, 1173.3, 1095.3, 976.5, 757.1, 577.2 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{30}H_{27}N_2O_4S_2$, 543.1406; found, 543.1414.

***N*-(Methyl(naphthalen-2-yl)(oxo)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4v).** Yield: 199.0 mg, 92%; white solid, mp: 78.5–80.5 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.26 (d, J = 1.3 Hz, 1H), 8.00 (d, J = 8.3 Hz, 1H), 7.85–7.82 (m, 2H), 7.70 (d, J = 8.3 Hz, 3H), 7.64–7.60 (m, 3H), 7.55 (t, J = 7.7 Hz, 2H), 7.32–7.28 (m, 1H), 7.20–7.16 (m, 1H), 7.03 (d, J = 8.1 Hz, 2H), 3.79 (s, 2H), 3.32 (s, 3H), 2.14 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.2, 144.8, 135.3, 135.2, 135.2, 135.1, 132.1, 131.1, 130.0, 129.8, 129.5, 129.4, 129.2, 127.9, 127.9, 126.7, 124.7, 124.7, 123.2, 121.3, 120.2, 117.4, 113.7, 44.1, 36.3, 21.4. IR (KBr): ν 3054.2, 2925.2, 1641.7, 1493.6, 1363.6, 1170.8, 1018.6, 939.8, 745.7, 575.9 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{28}H_{25}N_2O_4S_2$, 517.1250; found, 517.1256.

***N*-(Oxodiphenyl-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4w).** Yield: 183.8 mg, 87%; white solid, mp: 164–166 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.01 (d, J = 8.3 Hz, 1H), 7.76–7.71 (m, 6H), 7.60 (s, 1H), 7.56 (d, J = 7.8 Hz, 1H), 7.52–7.48 (m, 2H), 7.39–7.35 (m, 4H), 7.34–7.30 (m, 1H), 7.22–7.19 (m, 1H), 7.13 (d, J = 8.1 Hz, 2H), 3.81 (s, 2H), 2.26 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.7, 144.8, 139.4, 135.4, 135.2, 133.2, 131.1, 129.8, 129.4, 127.5, 126.8, 124.7, 124.6, 123.1, 120.3, 117.5, 113.6, 36.6, 21.5. IR (KBr): ν 3059.1, 2920.9, 1648.0, 1475.6, 1364.7, 1171.8, 1092.6, 994.6, 760.9, 592.1 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{29}H_{25}N_2O_4S_2$, 529.1250; found, 529.1260.

***N*-(Oxo(phenyl)(*p*-tolyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4x).** Yield: 190.8 mg, 88%; white solid, mp: 70–72 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.01 (d, J = 8.3 Hz, 1H), 7.74–7.69 (m, 4H), 7.61–7.59 (m, 3H), 7.55 (d, J = 7.8 Hz, 1H), 7.47–7.43 (m, 1H), 7.35–7.29 (m, 3H), 7.21–7.17 (m, 1H), 7.14 (d, J = 8.1 Hz, 2H), 7.08 (d, J = 8.4 Hz, 2H), 3.80 (s, 2H), 2.30 (s, 3H), 2.20 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.7, 144.7, 144.3, 139.7, 136.2, 135.4, 135.2, 133.1, 131.1, 130.1, 129.8, 129.3, 127.5, 127.4, 126.8, 124.6, 124.6, 123.1, 120.3, 117.5, 113.6, 36.6, 21.5. IR (KBr): ν 3062.3, 2922.7, 1649.7, 1446.8, 1364.9, 1172.0, 1094.4, 994.8, 811.9, 576.2 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{30}H_{27}N_2O_4S_2$, 543.1407; found, 543.1414.

***N*-(Bis(4-chlorophenyl)(oxo)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4y).** Yield: 188.3 mg, 79%; white solid, mp: 179–181 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.02 (d, J = 8.3 Hz, 1H), 7.76 (d, J = 8.4 Hz, 2H), 7.58–7.52 (m, 6H), 7.37–7.31 (m, 5H), 7.24–7.20 (m, 1H), 7.16 (d, J = 8.2 Hz, 2H), 3.79 (s, 2H), 2.29 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 178.6, 144.9, 140.3, 137.5, 135.4, 135.2, 130.9, 129.9, 129.8, 128.9, 126.8, 124.8, 124.6, 123.2, 120.3, 117.2, 113.7, 36.6, 21.5. IR (KBr): ν 3096.8, 2918.6, 1638.5, 1449.2, 1360.2, 1187.9, 1086.3, 989.7, 780.0, 578.2 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{29}H_{23}Cl_2N_2O_4S_2$, 597.0471; found, 597.0477.

***N*-((4-Benzoylphenyl)(oxo)(*p*-tolyl)-l6-sulfanylidene)-2-(1-tosyl-1*H*-indol-3-yl)acetamide (4z).** Yield: 191.2 mg, 74%; white solid, mp: 88.5–90.5 °C. Eluent: ethyl acetate/petroleum ether = 1 : 2 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.00 (d, J = 8.3 Hz, 1H), 7.79–7.70 (m, 8H), 7.65–7.61(m, 4H), 7.57 (d, J = 7.8 Hz, 1H), 7.51–7.47 (m, 2H), 7.34–7.30 (m, 1H), 7.23–7.19 (m, 3H), 7.14 (d, J = 8.1 Hz, 2H), 3.83 (d, J = 4.2 Hz, 2H), 2.38 (s, 3H), 2.26 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 195.0, 178.8, 144.9, 144.8, 143.1, 141.6, 136.3, 135.4, 135.2, 133.3, 131.0, 130.4, 130.3, 130.1, 129.8, 128.6, 127.7, 127.3, 126.8, 124.7, 124.6, 123.1, 120.3, 117.3, 113.6, 36.6, 21.5, 21.5. IR (KBr): ν 3060.4, 2922.4, 1657.4, 1447.3, 1366.6, 1120.8, 1094.0, 974.5, 811.6, 668.7 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{37}H_{31}N_2O_5S_2$, 647.1669; found, 647.1670.

General procedure for the synthesis of products 6

A screw capped Schlenk tube equipped with a stir bar was charged with *N*-allenyl-2-iodoaniline **5** (0.48 mmol), *N*H-sulfox-

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imine **3a** (0.40 mmol), $Pd(OAc)_2$ (0.02 mmol, 5 mol%), PPh_3 (0.08 mmol, 20 mol%), DBU (0.80 mmol), and dry toluene (2.0 mL). The reaction mixture was purged with a CO balloon 3 times, and then stirred at room temperature for 2–3 h. The reaction was monitored by TLC. Upon completion of the reaction, the reaction mixture was filtered through a short pad of silica gel and the solid residue was washed with ethyl acetate (~15 mL) unless otherwise noted. The concentration of the filtrate under reduced pressure provided the crude product, which was purified by silica gel column chromatography (petroleum ether/ethyl acetate = 10 : 1–3 : 1, v/v) to afford the desired compound **6**.

2-(1-Acetyl-1*H*-indol-3-yl)-*N*-(methyl(oxo)(phenyl)-l6-sulfanylidene)acetamide (6a). Yield: 119.0 mg, 84%; yellow oil. Eluent : ethyl acetate/petroleum ether = 1 : 5 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.42 (d, J = 6.9 Hz, 1H), 7.81 (d, J = 7.6 Hz, 2H), 7.62–7.56 (m, 2H), 7.48 (t, J = 7.7 Hz, 2H), 7.42 (s, 1H), 7.33 (t, J = 7.4 Hz, 1H), 7.27–7.23 (m, 1H), 3.79 (s, 2H), 3.28 (s, 3H), 2.54 (s, 3H).^{13}C NMR (100 MHz, $CDCl_3$) δ 179.5, 168.6, 138.4, 135.8, 133.9, 130.6, 129.6, 127.0, 125.2, 123.8, 123.5, 119.3, 116.8, 116.6, 44.1, 36.1, 23.9. IR (KBr): ν 3059.4, 2924.8, 1700.9, 1450.9, 1352.2, 1098.4, 978.2, 748.3, 655.9, 493.9 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{19}H_{19}N_2O_3S$, 355.1111; found, 355.1114.

***tert*-Butyl-3-(2-((methyl(oxo)(phenyl)-l6-sulfanylidene)amino)-2-oxoethyl)-1*H*-indole-1-carboxylate (6b).** Yield: 148.4 mg, 90%; yellow oil. Eluent : ethyl acetate/petroleum ether = 1 : 5 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.15 (d, J = 7.8 Hz, 1H), 7.81–7.79 (m, 2H), 7.61–7.56 (m, 3H), 7.47 (t, J = 7.8 Hz, 2H), 7.32–7.28 (m, 1H), 7.23–7.19 (m, 1H), 3.78 (s, 2H), 3.28 (s, 3H), 1.64 (s, 9H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.7, 149.7, 138.6, 135.5, 133.8, 130.6, 129.5, 127.1, 124.3, 124.3, 122.4, 119.6, 115.1, 115.0, 83.4, 44.0, 36.4, 28.2. IR (KBr): ν 2925.5, 1730.4, 1644.1, 1452.9, 13 669.0, 1223.2, 1086.4, 977.7, 746.2, 494.0 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{22}H_{25}N_2O_4S$, 413.1530; found, 413.1533.

***tert*-Butyl-3-(2-((methyl(naphthalen-2-yl)(oxo)-l6-sulfanylidene)amino)-2-oxoethyl)-1*H*-indole-1-carboxylate (6c).** Yield: 160.8 mg, 87%; yellow liquid. Eluent : ethyl acetate/petroleum ether = 1 : 5 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.33 (d, J = 1.5 Hz, 1H), 8.19 (d, J = 7.8 Hz, 1H), 7.88 (t, J = 9.3 Hz, 2H), 7.77 (d, J = 8.0 Hz, 1H), 7.71–7.68 (m, 1H), 7.66–7.61 (m, 2H), 7.59–7.55 (m, 2H), 7.35–7.31 (m, 1H), 7.24–7.20 (m, 1H), 3.82 (s, 2H), 3.35 (s, 3H), 1.64 (s, 9H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.8, 149.7, 135.3, 132.2, 130.7, 130.0, 129.4, 129.4, 129.2, 127.9, 127.8, 124.4, 124.4, 122.5, 121.5, 119.7, 115.2, 115.2, 83.4, 44.1, 36.5, 28.2. IR (KBr): ν 3055.4, 2929.2, 1730.6, 1643.9, 1453.5, 1309.0, 1083.6, 858.6, 747.4, 479.0 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{26}H_{27}N_2O_4S$, 463.1686; found, 463.1696.

***tert*-Butyl-3-(2-((methyl(oxo)(*o*-tolyl)-l6-sulfanylidene)amino)-2-oxoethyl)-1*H*-indole-1-carboxylate (6d).** Yield: 158.5 mg, 93%; yellow oil. Eluent : ethyl acetate/petroleum ether = 1 : 5 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.13 (d, J = 8.0 Hz, 1H), 7.97–7.94 (m, 1H), 7.58–7.55 (m, 2H), 7.49–7.45 (m, 1H), 7.33–7.29 (m, 2H), 7.25 (d, J = 4.9 Hz, 1H), 7.21–7.17 (m, 1H), 3.77 (d, J = 0.8 Hz, 2H), 3.31 (s, 3H), 2.47 (s, 3H), 1.64 (s, 9H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.4, 149.7, 137.0, 136.6, 135.5, 133.7, 133.2, 130.6, 129.1, 126.9, 124.3, 124.3, 122.4, 119.5, 115.1, 114.9, 83.4, 42.9, 36.3, 28.2, 20.1. IR (KBr): ν 2978.6, 1730.7, 1643.7, 1453.7, 1369.4, 1157.6, 1084.6, 975.8, 756.8, 462.6 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{23}H_{27}N_2O_4S$, 427.1686; found, 427.1693.

***tert*-Butyl-3-(2-(((4-chlorophenyl)(methyl)(oxo)-l6-sulfanylidene)amino)-2-oxoethyl)-1*H*-indole-1-carboxylate (6e).** Yield: 144.5 mg, 81%; yellow oil. Eluent : ethyl acetate/petroleum ether = 1 : 5 ^{1}H NMR (400 MHz, $CDCl_3$) δ 8.18–8.15 (m, 1H), 8.11 (d, J = 7.9 Hz, 1H), 7.56–7.51 (m, 3H), 7.49–7.43 (m, 2H), 7.29–7.26 (m, 1H), 7.21–7.17 (m, 1H), 3.77 (d, J = 0.9 Hz, 2H), 3.43 (s, 3H), 1.64 (s, 9H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 179.0, 149.7, 134.7, 132.1, 131.6, 131.1, 130.7, 127.8, 124.4, 124.2, 122.4, 119.5, 115.1, 114.7, 83.3, 42.1, 35.8, 28.2. IR (KBr): ν 2926.4, 1645.1, 1447.8, 1365.1, 1173.0, 1094.2, 977.0, 814.6, 670.7, 538.0 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{22}H_{24}ClN_2O_4S$, 447.1145; found, 447.1140.

General procedure for the gram-scale synthesis of 4a

A screw capped Schlenk tube equipped with a stir bar was charged with *N*-(2-iodophenyl)-4-methyl-*N*-(propa-1,2-dien-1-yl)benzenesulfonamide **7a** (987.0 mg, 2.4 mmol), *N*H-sulfoximine **3a** (310 mg, 2.0 mmol), $Pd(OAc)_2$ (0.1 mmol, 5 mol%), PPh_3 (0.4 mmol, 20 mol%), DBU (4.0 mmol), and dry toluene (10 mL). The reaction mixture was purged with a CO balloon 3 times, and then stirred at room temperature for 6 h. The reaction was monitored by TLC. Upon completion of the reaction, the reaction mixture was filtered through a short pad of silica gel and the solid residue was washed with ethyl acetate (~15 mL). The concentration of the filtrate under reduced pressure provided the crude product, which was purified by silica gel column chromatography (petroleum ether/ethyl acetate = 10 : 1–3 : 1, v/v) to afford the desired compound **4a**, 801 mg, 86% yield.

General procedure for the preparation of compound 8

To an ice cold stirred solution of β-indolyl sulfoximidoyl amide **4a** (0.3 mmol) in freshly distilled CH_2Cl_2 (3.0 mL) was added dropwise the borane complex (3.0 equiv.) under a N_2 atmosphere. After stirring for 3 h at 0 °C, the ice bath was removed and the mixture was warmed to r.t. The reaction mixture was stirred at room temperature overnight. Upon completion of the reaction as monitored by TLC, the reaction was carefully quenched with MeOH and H_2O, and the resulting mixture was extracted with CH_2Cl_2. The combined organic extracts were dried over $MgSO_4$, and the solvent was evaporated under reduced pressure. The crude residue was purified through silica gel chromatography to yield the pure product **7** as a white solid in 65% yield.

Methyl(phenyl)((2-(1-tosyl-1*H*-indol-3-yl)ethyl)imino)-l6-sulfanone (8). Yield: 117.5 mg, 65%; white solid, mp: 103–104 °C. Eluent : ethyl acetate/petroleum ether = 1 : 5 ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.95 (d, J = 8.3 Hz, 1H), 7.76–7.72 (m, 4H), 7.57–7.54 (m, 1H), 7.47–7.43 (m, 2H), 7.39 (d, J = 7.8 Hz, 1H), 7.35 (s, 1H), 7.28–7.25 (m, 1H), 7.17–7.13 (m, 3H), 3.31–3.24

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(m, 1H), 3.10–3.03 (m, 4H), 2.94–2.90 (m, 2H), 2.29 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 144.7, 139.2, 135.4, 135.2, 132.9, 131.2, 129.8, 129.4, 128.5, 126.8, 124.5, 123.3, 122.9, 121.2, 119.6, 113.6, 45.1, 43.5, 28.5, 21.5. IR (KBr): ν 2922.4, 1446.6, 1366.8, 1173.0, 980.2.6, 745.4, 690.6, 575.1, 516.9 cm^{-1}; HRMS (ESI) m/z: $[M + H]^+$ calcd for $C_{24}H_{25}N_2O_3S_2$, 453.1301; found, 453.1312.

Conflicts of interest

There are no conflicts of interest to declare.

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